

Chapter 3

Differential Equations

3.1 DEFINITION

A differential equation involving derivatives of one or more dependent variables with respect to one or more independent variable is called a differential equation (DE).

There are two types of differential equations:

(1) Ordinary Differential Equation:

A differential equation involving derivatives with respect to a single independent variable is called an ordinary differential equation (ODE).

(2) Partial Differential Equation:

A differential equation involving partial derivatives with respect to more than one independent variable is called a partial differential equation (PDE).

3.2 ORDER AND DEGREE OF A DIFFERENTIAL EQUATION

The Degree of a Differential equation is the degree of the highest derivative occurring in the Differential equation.